Solution Requirements

Date	08 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what the solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Not required for the current prototype; the web dashboard is openly accessible to anyone with the LocalTunnel link.	Low
2	Authorization levels	Single-role access: any visitor to the dashboard can submit indicator values and request a prediction; no admin/user distinction in this version.	Low
3	External interfaces	Dataset sourced from the Kaggle Human Development Index (Full) dataset; no live third-party API integration in the current version.	Medium
4	Transactions processing	Handling input of Life Expectancy, Expected Years of Schooling, and GNI per capita, and returning a predicted HDI score.	High
5	Reporting	Dashboard displays the predicted HDI score along with a colour-coded development-tier badge (Low / Medium / High / Very High).	High
6	Business rules, etc.	System must validate that input values fall within historically realistic ranges (e.g., GNI per capita > 0, schooling ≥ 0 years).	High
7	Compliance to laws or regulations	No personally identifiable information is collected or stored; only aggregate, publicly sourced socio-economic indicator values are processed.	High
8	Other	Reproducible inference through serialized model	Medium

S.No	Requirement Category	Requirement Description	Priority
		weights (.pt) and saved feature-scaling parameters.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	API response time for HDI prediction must be near-instantaneous after form submission.	Loads in < 2 seconds
2	Scalability	GPU-based preprocessing (cuDF) should scale to multi-year datasets without excessive slowdown.	Handles full multi-year CSV dataset
3	Security & Data Privacy	No personal data is collected; only aggregate socio-economic indicator values are transmitted and processed.	No PII stored or transmitted
4	Reliability & Availability	The Flask server and LocalTunnel URL should remain stable for the duration of the notebook / demo session.	Stable for full demo session
5	Usability & Accessibility	Web form should be a simple, single-page layout with clear numeric input fields and validation ranges.	Intuitive, no installation required
6	Other	Codebase (model, training script, Flask app) should be modular and documented for easy updates.	Maintainable / reusable functions